Week11 (11/27-12/3)

MoonWalker Weekly Meeting Slides

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MoonWalker Weekly Meeting

11/27 - 12/3 Summary

Elijah, Rowel, Martin

Progress Update

- Martin
  - Implemented Unity-hosting toggle upon collision
  - Create json object for collision box information
- Elijah
  - Implemented Consumer-Producer script with:
    - Physics Parameters extraction
    - Collision detection-based switching
    - Switching is based on the Hosting Flag from the JSON file
- Rowel
  - Update API to include Physics Parameters
    - Update send-message to allow physics in messages
  - Continue updating HSML Schema (hsml.jsonID) and Schema Properties.xlsx
    - New Parameters (Inertia, center of mass)

Producer to Consumer Switching - Collision Detection

Omniverse Physics Parameters Extraction

- Parameters Extracted:
  - Mass, Inertia, Center of Mass
- Challenge:
  - Not really for the script itself, but in the Omniverse world the Cadre was not given the “Mass” physics property.
  - Fix: Simply gave it one, but the values are “Autocomputed” which returns:
    - Mass: 0.0
    - Inertia: 0.0, 0.0, 0.0
    - Center of Mass: -inf, -inf, -inf
  - Script still extracts the parameters


```
{
    "position": [
        4.009741306304932,
        -6.284987926483154,
        -0.29608139395713806
    ],
    "rotation": [
        0.5036052319019186,
        -0.4292423767077468,
        -0.46695992424280014,
        0.586584334593465
    ],
    "mass": 0.0,
    "inertia": {
        "type": "diagonal",
        "values": [
            0.0,
            0.0,
            0.0
        ]
    },
    "center_of_mass": [
        -Infinity,
        -Infinity,
        -Infinity
    ],
    "hosting": false,
    "modifiedDate": "2025-12-02T20:26:45.386105"
}
```

```
moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing Li/Desktop/MoonWalker$ curl -X POST \
  "http://localhost:8000/producer/send-message?topic=test-topic" \
  -H "Content-Type: application/json" \
  -d '{
    "@context": "https://digital-twin-interoperability.github.io/hsml-schema-context/hsml.jsonld",
    "@type": "Agent",
    "name": "Viper_A",
    "swid": "did:key:generateBeforeDemoWithRegistration",
    "url": "1GBcmwJh2rQ4CSR3_CiDyF6oVdOxz6hLG",
    "creator": {
      "@type": "Person",
      "name": "Jared Carrillo",
      "swid": "did:key:personMustBeRegisteredBefore"
    },
    "dateCreated": "2025-12-03",
    "dateModified": "2025-12-03T12:00:00",
    "encodingFormat": "application/x-obj",
    "contentUrl": "https://example.com/models/3dmodel-001.obj",
    "description": "Unity simulation agent HSML",
    "platform": "Unity",
    "inControl": true,
    "spaceLocation": [
      {
        "@type": "Hyperspace",
        "name": "Moon"
      }
    ],
    "position": [
      { "@type": "schema:PropertyValue", "name": "xCoordinate", "value": 0.0 },
  }'] { "@type": "schema:PropertyValue", "name": "scale", "value": 100 } 1.2 },
{"message":"Message sent successfully"}moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing Li/Desktop/M
moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing Li/Desktop/MoonWalker$
```

Verified Old Messages are still compatible

```
HSML-API   MySQL   Kafka   Ubuntu
  -F "private_key=@csun.pem"
{"detail":"Authentication failed."}moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing Li/D
moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing Li/Desktop/MoonWalker$
moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing Li/Desktop/MoonWalker$ curl -X POST \
  "http://localhost:8000/producer/send-message?topic=test-topic" \
  -H "Content-Type: application/json" \
  -d '{
    "position": [1.0, 2.0, 3.0],
    "rotation": [1.0, 0.0, 0.0, 0.0],
    "mass": 120.5,
    "inertia": {
      "type": "diagonal",
      "values": [5.0, 5.0, 5.0]
    },
    "center_of_mass": [0.1, 0.2, 0.3],
    "hosting": true,
    "modifiedDate": "2025-12-03T10:00:00.000000"
  }'
{"message":"Message sent successfully"}moonwalker@DESKTOP-TOD84J7:/mnt/c/Users/Bingbing
```

API accepts messages with physics

Unity: Ran into Licensing Issue

mykey.pem | private_key_ | hsml.json | cadreA_omn | Elijah_cadre/ | forced_editc | UnityEntitler | Unity_lic.ulf | unity_log.txt | unity_

File Edit View H1

```
[LicensingClient] Error: Code 10 while verifying Licensing Client signature (process Id: 24552, path: "C:/Program Files/Unity
Hub/UnityLicensingClient_V1/Unity.Licensing.Client.exe")
[Licensing::Module] Error: LicensingClient has failed validation
[LicensingClient] Channel doesn't exist: "LicenseClient-a_h-2021.3.1"
[Licensing::Module] Successfully launched the LicensingClient (PId: 15120)
[LicensingClient] Handshaking with LicensingClient (version: 1.9.0+249add7)
[Licensing::Module] Successfully connected to LicensingClient on channel: "LicenseClient-a_h-2021.3.1" (connect: 0.42s, validation: 0.04s,
handshake: 0.61s)
[Licensing::Module] Connected to LicensingClient (PId: 15120, launch time: 0.00, total connection time: 1.07s)
Entitlement-based licensing initiated
[Licensing::Module] Error: Access token is unavailable
[LicensingClient] Licenses updated successfully

LICENSE SYSTEM [20251126 13:4:2] Next license update check is after 2025-11-27T20:56:07


LICENSE SYSTEM [20251126 13:4:2] Machine binding 5 mismatch: (value of current machine) 0c:15:a3:b6:7c:6a != 48:e7:da:2b:04:3b (value in
license file). Reason might be there is a hardware change on this machine after the license was activated.


LICENSE SYSTEM [20251126 13:4:2] Machine identification is invalid for current license.
Current license is invalid and cannot be activated. You must delete the license file and then activate a new license on current machine.
Exiting without the bug reporter. Application will terminate with return code 0
```

Ln 3, Col 70 | 1,519 characters | Plain text | 100% | Windows (CRLF) | UTF-8

In order to recreate collision with one-host:

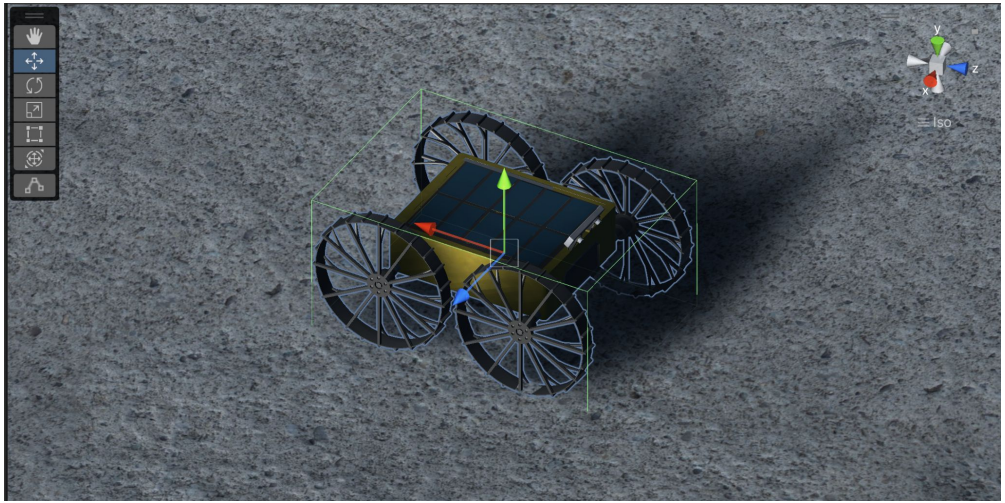
- Hosting simulation needs two things:
  - Physics Parameters ✅
  - Collision box ❌

Why not just physics?

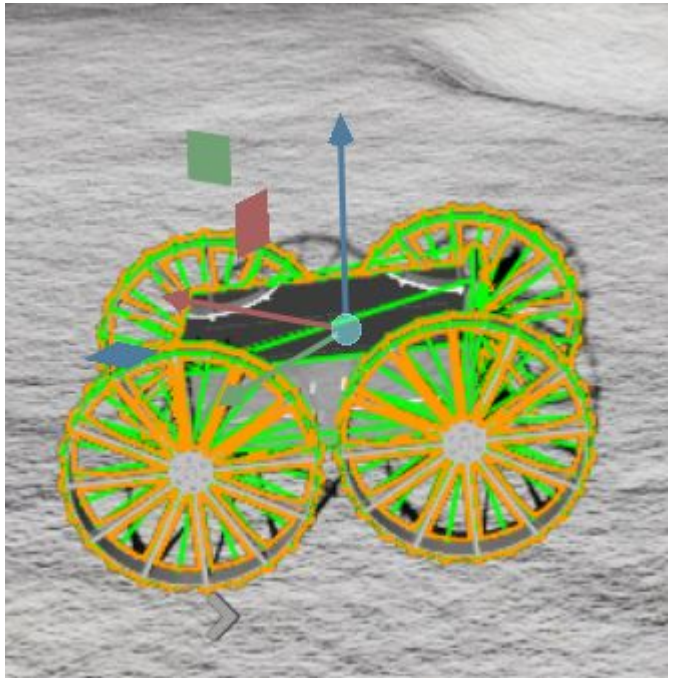
- Host sim. needs to know ghost's collision boundaries in order to correctly use the physics parameters
- Despite the mesh, the simulation doesn't "understand" its shape
- Currently, colliders are different in both simulations

Cadre Collider: Unity vs Omniverse

Box Collider

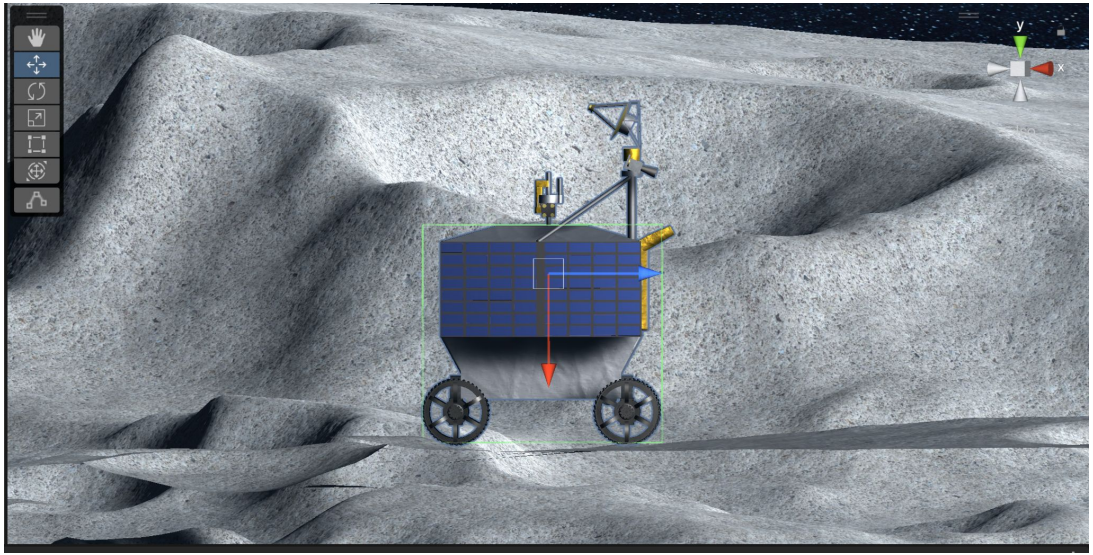


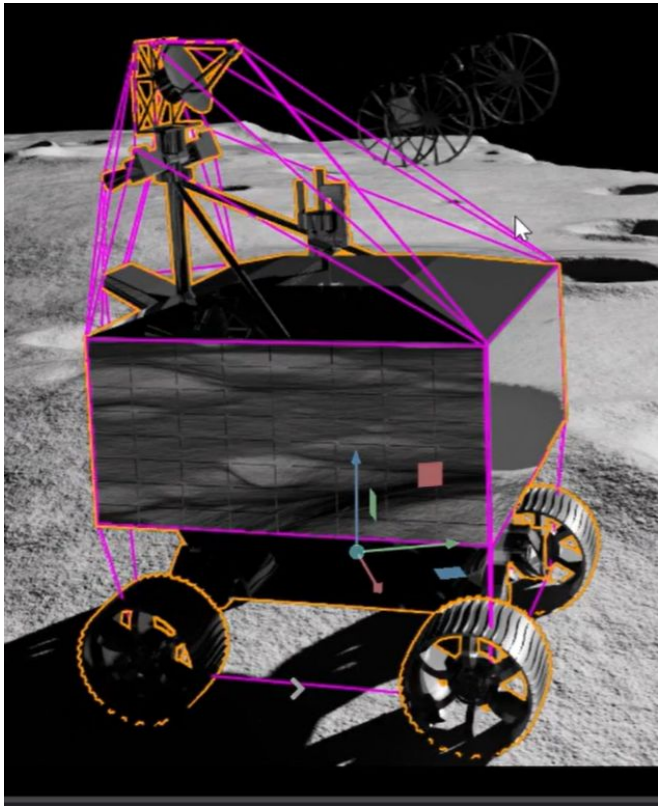
Convex Hull



Viper Collider: Unity vs Omniverse

Box Collider



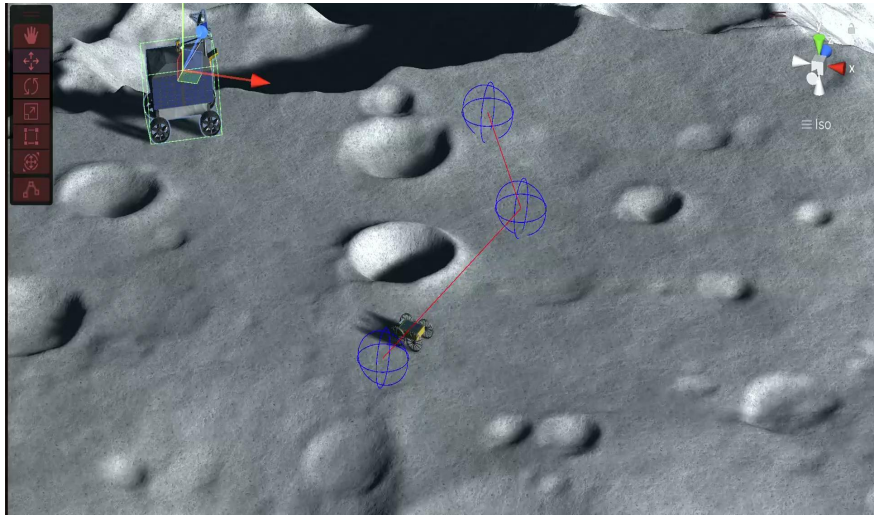


Convex Hull

Alongside physics params, send collider shape

```
{
    "colliders", new JArray {
    new JObject {
        { "@type", "Collider" },
        { "shape", "Box" },
        { "center", new JObject {
            { "x", -0.42076 },
            { "y", -0.01154 },
            { "z", 0.030452 }
        }},
        { "size", new JObject {
            { "x", 1.626567 },
            { "y", 1.531231 },
            { "z", 1.788746 }
        }}
    }
}}
```

Unity Local Collision Testing



To use as reference for Unity-hosting

Overall:

- We can toggle between both host simulations
- Sending physics param. to API works

To-do:

- Document Unity license issue in case someone encounters it in the future
- Send collision box information
- Set timer or similar to stop one-host, go back to normal interoperability

Action Plan

- Rowel
  - Finish Updating API (Confirm compatibility with Unity Scripts)
    - Possibly adding more parameters
  - Begin testing
- Elijah
  - Work with Rowel & Martin on implementing hsml schema changes
    - Based on JSON files being written by Omniverse
  - Incorporate other Physics Parameters in the JSON files
  - Generalized the Consumer-Producer script written for Cadre
- Martin
  - Work with Rowel & Elijah on implementing collision box parameters.